OLTP

- OLTP, i.e. **Online Transaction Processing**, is a type of data processing that consists of executing a number of transactions occurring concurrently—**online banking, shopping, order entry, or sending text messages**, for example.
- The majority of OLTP systems are meant to manage numerous short atomic operations that keep databases in line.
- To maintain transaction integrity and reliability, these systems support **ACID** (Atomicity, Consistency, Isolation, Durability) properties.
- An example considered for OLTP System is ATM Center a person who authenticates first will receive the amount first and the condition is that the amount to be withdrawn must be present in the ATM.

OLTP:

➢Working of OLTP:

- OLTP works by processing real-time transactions through a structured three-tier architecture that ensures speed, consistency, and reliability.

Transaction Flow

- A transaction starts in the **presentation tier** (user interface like web apps), where inputs like a bank transfer request are captured.
- It moves to the **business logic tier** (app servers), which validates rules, checks balances, and prepares database operations.
- Finally, the **data tier** (RDBMS) executes atomic changes with ACID guarantees—if any step fails, it rolls back completely.

OLTP:

➢**Step-by-Step Process**

1. Input Capture: User submits data (e.g., order details).
2. Validation: Logic tier verifies integrity and availability.
3. Execution: Database applies locks/indexes for concurrent writes.
4. Commit/Rollback: Success commits changes; failure reverts.
5. Response: Confirmation sent back to user in milliseconds.

OLTP:

➢**Benefits of OLTP Services**

- Allow users to quickly read, write and delete data operations.
- Support an increase in users and transactions for real-time data access.
- Provide better data protection through multiple security features.
- Aid in decision-making with accurate, up-to-date data.
- Ensure data integrity, consistency and high availability.

OLTP:

➢**Drawbacks of OLTP Services**

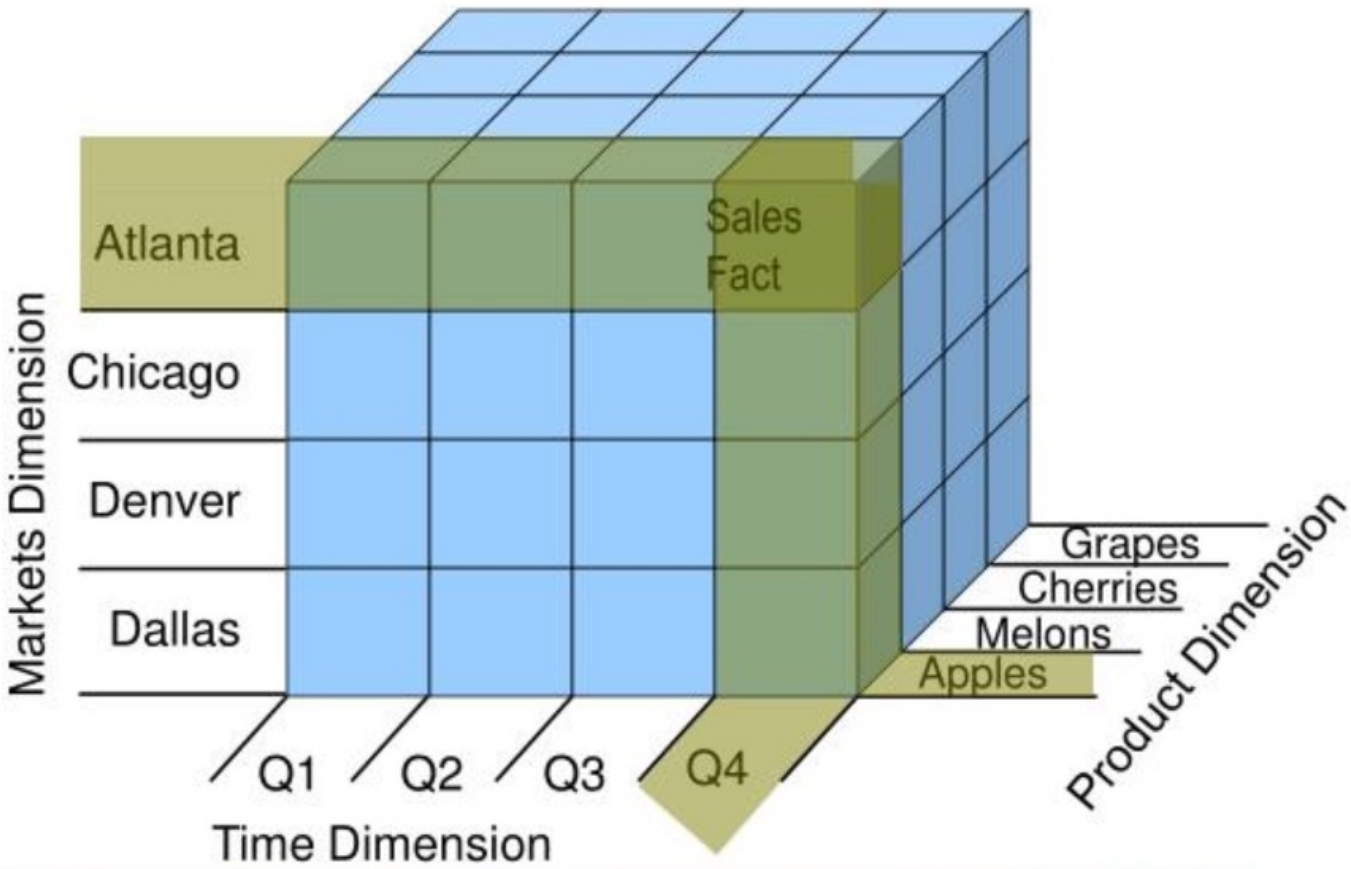
- Limited analysis capability, not suited for complex analysis or reporting.
- High maintenance costs due to frequent updates, backups and recovery.
- Susceptible to disruption during hardware failures, impacting online transactions.
- Prone to issues like duplicate or inconsistent data.

OLAP:

- Online Analytical Processing (OLAP) refers to software tools used for the analysis of data in business decision-making processes.
- It is technology for performing high-speed complex queries or multidimensional analysis on large volumes of data in a data warehouse, data lake or other data repository.
- OLAP systems generally allow users to extract and view data from various perspectives, many times they do this in a multidimensional format which is necessary for understanding complex interrelations in the data.
- Any type of Data Warehouse System is an OLAP system. Examples; Spotify personalizes homepages with custom songs and playlists based on user preferences or Netflix movie recommendation system.

OLAP:

- Most business data have multiple dimensions—multiple categories into which the data are broken down for presentation, tracking, or analysis.
- For example, sales figures might have several dimensions related to location (region, country, state/province, store), time (year, month, week, day), product (clothing, men/women/children, brand, type), and more.
- But in a data warehouse or data lake, data sets are stored in tables, each of which can organize data into just two of these dimensions at a time.
- OLAP extracts data from multiple relational data sets and reorganizes it into a multidimensional format that enables very fast processing and very insightful analysis.


Atlanta
Chicago
Denver
Dallas
Markets Dimension
Sales
Fact
Q1
Q2
Q3
Q4
Time Dimension
Grapes
Cherries
Melons
Apples
Product Dimension

OLAP:

➢ Benefits of OLAP Services

OLAP offers several advantages that make it a valuable technology for data analysis and business intelligence. Here are the main advantages of OLAP:

- **Interactive Analysis**: OLAP provides an interactive environment for users to explore data from various dimensions and perspectives. Users can drill down, roll up, slice, dice, and pivot data to gain insights and answer specific analytical questions.
- **Quick Decision-Making**: With fast query response times and interactive capabilities, OLAP empowers users to make quicker and more informed decisions. This is especially important in fast-paced business environments.
- **Flexibility**: OLAP allows users to create ad-hoc queries and modify queries on the fly to address new questions as they arise. This flexibility is crucial for exploring unexpected trends and patterns.

OLAP:

- **Reduced Data Complexity**: OLAP systems abstract the complexity of underlying data structures by providing a user-friendly interface. Users can focus on exploring insights rather than dealing with complex database queries.
- **Enhanced Collaboration**: OLAP's interactive nature facilitates collaboration among team members. Different users can explore the same data from their own perspectives and contribute insights to discussions.

OLAP:

➢Drawbacks of OLAP Services

- Requires professionals to handle the data because of its complex modeling procedure.
- Expensive to implement and maintain in cases when datasets are large.
- Data analysis occurs only after extraction and transformation, leading to system delays.

OLAP vs OLTP:

- Difference between OLAP and OLTP

Category	OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
Data Source	Historical data from multiple databases.	Current operational data.
Purpose	Used for analysis and decision-making.	Used for day-to-day transactions.
Method Used	Uses a data warehouse.	Uses a standard DBMS.
Normalization	Tables are not normalized.	Tables are normalized (3NF).
Query Type	Complex, read-heavy queries (slow).	Simple, read/write queries (fast).

OLAP vs OLTP:

Category	OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
Data Volume	Large (TB–PB).	Small (MB–GB).
Update Frequency	Updated periodically in batches.	Updated frequently by users.
Backup & Recovery	Periodic backup.	Continuous and rigorous backup.
Users	Used by analysts, managers and executives.	Used by clerks and operational staff.
Focus	Subject-oriented (analysis-focused).	Application-oriented (operation-focused).

OLAP Architectures:

There are two main architectural approaches in OLAP: Multidimensional (MOLAP) and Relational (ROLAP).

➢ **Multidimensional OLAP (MOLAP) Architecture:**

MOLAP architecture is designed around the concept of multidimensional cubes. These cubes store pre-aggregated data based on various dimensions, enabling fast query response times.

The architecture involves the following components:

Cubes: The core element of MOLAP, cubes are multidimensional structures that store data in cells at intersections of dimensions. Each cell contains aggregated data or measures.

Dimensions: Dimensions are the various perspectives or attributes by which data can be analyzed. Common examples include time, geography, product categories, etc.

OLAP:

- **Measures**: Measures are the data values that are being analyzed, such as sales revenue, profit, quantities sold, etc. Measures are stored in the cells of the cube and can be aggregated across dimensions.
- **Aggregations**: MOLAP systems pre-calculate aggregations during data processing and store them in the cube. This speeds up query response times because calculations are performed in advance.
- **Calculation Engine**: The calculation engine of the MOLAP system handles complex calculations, aggregations, and formula-based operations on the data.
- **Storage**: Data in MOLAP is stored in proprietary formats optimized for fast retrieval. This storage structure contributes to the quick query performance of MOLAP systems.
- **User Interface**: MOLAP systems provide user-friendly interfaces that allow users to interact with multidimensional data, performing operations like slicing, dicing, drilling, and pivoting.

OLAP:

➢Relational OLAP (ROLAP) Architecture

ROLAP architecture utilizes relational databases as the backend for data storage and processing. It involves the following components:

- **Relational Database**: Data is stored in relational tables within a database. Each table contains dimensions, measures, and the relationships between them.
- **Metadata**: Metadata describes the relationships between dimensions, measures, and other elements. It's used to generate SQL queries that retrieve and combine data for analysis.
- **Dimension Tables**: These tables store the attributes and hierarchies of dimensions. Each row in a dimension table represents a unique dimension value.

OLAP:

Fact Tables: Fact tables store the measures and foreign keys that connect to dimension tables. Fact tables contain the data that is being analyzed.

SQL Engine: ROLAP systems use SQL queries to retrieve data from the relational database based on user requests. These queries can involve complex joins and calculations.

Aggregations (Optional): Similar to MOLAP, ROLAP systems can employ pre-calculated aggregations to improve query speed.

User Interface: Users can construct and execute SQL queries, see results, and produce reports using the interfaces provided by ROLAP systems.

OLAP:

➢Hybrid OLAP:

Hybrid online analytical processing (HOLAP) combines MOLAP and ROLAP to provide the best of both architectures.

HOLAP allows data engineers to quickly retrieve analytical results from a data cube and extract detailed information from relational databases.